What VeReMi Measures: Structural Artefacts in VANET Misbehaviour Detection Benchmarks

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Abstract—One untrained scalar solves six of eight VeReMi archives. It reads message timing alone, never a claimed position. These archives are the field's main shared misbehaviour benchmark. Reported detection rates on them sit between 94% and 99.9%. We audit them by measurement: every claim is re-derived from source. We examine the eight Sybil archives of the extension. Pseudonyms encode the true sender in 86.5–100% of identity pairs. No benign vehicle in them emits a second pseudonym. Attacker prevalence is 30.0% of vehicles in all eight. Attacker identities form 68–98% of the identity population. Ground-truth files list up to 28 times fewer identities than receivers observe. We then run one detector through seven progressively stricter protocols. Removing ground-truth aggregation is associated with a 0.060 fall. We evaluate the pseudonym feature in its original representation. The lowest stable evaluable prevalence is 3% dense, 5% sparse. Ranking metrics barely move with prevalence; precision metrics collapse. We show VeReMi 2018 cannot validate signal-strength position verification. Its measured path-loss exponent is 0.93 against a free-space two. That detector then reaches AUC 0.530, barely above chance. We release code, controls and a reporting checklist for future work.

Index Terms—VANET, misbehaviour detection, Sybil attack, benchmarking, reproducibility, VeReMi.

I. Introduction

A Sybil attacker presents many identities from one radio. In vehicular networks the consequence is direct and physical. Safety applications count reporters before believing a claim. Fake congestion, forged warnings and vote manipulation all follow. Douceur formalised the problem for peer-to-peer systems [1]. Newsome adapted the defence taxonomy to sensor networks [2]. Vehicular work inherited both, then grew for twenty years.

Progress is normally judged by reported detection rates. Those rates have risen steadily and now approach saturation. Early trajectory schemes reported roughly 98% detection [9]. Recent deep models report between 94% and 99.9%. Such numbers imply the problem is essentially solved. Deployment experience does not support that conclusion.

Security machine learning has met this pattern before. Sommer and Paxson warned about it fifteen years ago [8]. Arp et al. catalogued the pitfalls that produce it [24]. Sampling bias and spurious correlation head their list. We found that lens rarely applied to vehicular benchmarks.

Comparability is the missing ingredient, and it is scarce. VeReMi introduced a common reference for misbehaviour detection [16]. Its extension added attack families including explicit Sybil variants [19]. Shared evaluation is nonetheless rare in this literature. Of our 92 full texts, 5 mention VeReMi at all. Only 7 name any public dataset whatsoever. The remaining 92% name no public dataset at all. So these archives are not ubiquitous; they are the main alternative. That is precisely why their contents matter.

We therefore ask what these archives actually contain. We parse them and quantify their structure directly. We then re-derive headline results under explicit controls. Our contributions are as follows.

- We document six structural artefacts across all eight Sybil archives. Each is measured from source, not read from documentation.
- We show a single untrained scalar solves most archives. No training, no linkage and no physical reasoning are required.
- We separate timing evidence from geometric evidence. Timing alone reaches AUC near unity on several families.
- We run one detector through seven progressively stricter protocols. Each stage measures the change associated with one removed control.
- We sweep prevalence, jitter, attack intensity and beacon rate.
- We show VeReMi 2018 cannot validate signal-strength verification.
- We release controls, scripts and a reporting checklist.

Our stance is deliberately narrow. We do not accuse any author of exploiting a loophole. The published protocols we reconstruct were reasonable when written. The defensible claim is simpler and more uncomfortable. These benchmarks substantially simplify the intended deployment task.

II. Background and Related Work

A. Sybil detection in vehicular networks

Early vehicular defences used physics rather than cryptography. Xiao et al. verified claimed positions from signal-strength distributions [3]. Guette and Ducourthial then quantified the attacker's freedom [4]. Transmit power tuning and directional antennas defeat naive checks. Bouassida et al. measured signal variation against real hardware [6].

Infrastructure-anchored designs followed. Park et al. used roadside timestamp series instead of certificates [7]. Footprint made anonymous roadside proofs into a trajectory identity [9]. P2DAP detected pseudonym reuse while preserving anonymity [5], [10]. Later work extended these ideas to conspiring attackers [14].

Statistical and learning approaches then dominated. Voiceprint compared signal-strength series without a propagation model [15]. Ayaida et al. detected contradictions against a traffic-flow model [18]. Dutta and Chellappan clustered platoon dispersion over time [11]. Grover et al. combined several verification variables [13]. Yu et al. consolidated the position-verification line [12]. Recent systems use ensembles, graphs and gradient boosting [23], [33].

A parallel line prices identities instead of detecting them. Baza et al. required proofs of work and location [20]. Khatri et al. anchored location proofs on a blockchain [28]. Hadri et al. used verifiable delay functions with fog nodes [36]. Credential designs prevent concurrency outright [21], [26]. Fog and 5G variants target lightweight deployment [27].

Surveys cover this ground thoroughly. Boualouache and Engel survey machine-learning misbehaviour detection [25]. A recent survey covers federated and edge approaches [34]. We therefore do not attempt another survey.

B. What our critique would touch

Azam et al. aggregate movement features per node, then apply random folds. That is exactly the first protocol in our cascade. Guven and Tayşi built a new dataset for related reasons [29]. Their motivation and ours agree on the substrate problem.

C. Benchmarks and their scrutiny

VeReMi was created to make results comparable [16]. Its extension added Sybil families and more scenarios [19]. Evaluation practice around them is rarely examined.

A newer release has since appeared [35]. VeReMi NextGen moves to the InTAS scenario and adds attacks. It ships predefined training and validation splits. Its aliases do not encode the sender: at most 0.02%. That single change removes our first structural artefact. Our audit concerns VeReMi and its extension only. We have not audited NextGen, and it deserves the same treatment.

Table I places the four artefacts side by side. It also states what this audit adds to each.

Only the Guven set pairs Sybil attacks with received power. We could not evaluate it, for the reason given later. We found no study in our search that measures them structurally.

Adjacent work does question evaluation. SixPack evades detectors by abusing anti-lock braking signals [22]. That work targets detectors, not the dataset itself. Iwendi et al. illustrate the reported-accuracy pattern we examine [17]. We found no study in our search that measures these archives structurally.

TABLE I
The four public artefacts, and what we add. Sybil means explicit Sybil families, not misbehaviour generally.

Artefact	Sybil	RSSI	Rotation	What we add
VeReMi [16]	no	yes	no	path loss and detector
Extension [19]	yes	no	no	artefacts, cascade, sweeps
Guven [29]	yes	yes	—	availability status only
NextGen [35]	yes	no	no	first structural probe

D. Why realism matters

Attack impact depends strongly on attacker share. Söderhäll et al. measured navigation-service manipulation [30]. Sybils at 3% of users raised average travel time 20%. Our search found no field measurement of deployed V2X prevalence. Neither 3% nor 30% is an observed field figure. The honest response is a sweep, which we report. Physical-layer methods stay popular despite substrate limits [31], [32].

III. Method

A. Corpus

We assembled 331 records on vehicular Sybil attacks. Sources were OpenAlex, Semantic Scholar, arXiv, OpenAIRE and HAL. The index was built on 3 September 2026. Records span publication years 2006 to 2026.

Screening kept work on Sybil attacks in vehicular networks. We deduplicated on digital object identifier and on title. Records without a reachable open-access file were kept for counting only. Ninety-two open-access full texts were obtained and parsed. We extracted reported experimental parameters by pattern matching.

The flow is therefore 331 records screened, 92 full texts analysed. We report it in that form rather than as a systematic review. Two requirements of such a review are not met here. The exact query strings were not preserved during collection. Screening was done by one author without a second rater. We state both gaps rather than describe the search as systematic.

The corpus supplies context, not the paper's main claims. It is a convenience sample, not an exhaustive review. Statements below about absent prior work refer to this search.

B. Datasets and their schema

We used three public artefacts, verified by MD5 against Zenodo. VeReMi 2018 provides received power and position falsification attacks. Its record is 10.5281/zenodo.20081895. VeReMi Extension provides Sybil families and pseudonyms. Its record is 10.5281/zenodo.20090854. A preprocessed derivative provides ready-to-use feature rows. Its record is 10.5281/zenodo.14903687. All three are CC-BY licensed. We name the records because titles alone are ambiguous.

TABLE II
Fields read, and who may see them.

Field	Meaning	Detector sees
type	record kind, 2/3/4	yes
rcvTime	reception time	yes
sendTime	claimed send time	yes
messageID	transmission identifier	yes
pos, spd, acl, hed	claimed kinematics, each with its noise variance	yes
RSSI	received power, 2018 only	yes
receiver id	from the log filename	metadata
senderPseudo	raw pseudonym	hashed first
sender	true vehicle identifier	no
An in filename	attack type label	no

Table II lists every field we read. The third column is our central methodological device. A field is public if a receiver could observe it. Everything else is evaluation-only and never reaches a model.

C. Preprocessing

The pipeline has five steps and no hidden state.

1) Parse every receiver log in every simulation window.
2) Replace each pseudonym with BLAKE2b(salt, archive, pseudonym).
3) Deduplicate on messageID within a window.
4) Build features per identity, or per identity pair.
5) Split by true vehicle, then fit and score.

Step three matters more than it looks. Each transmission is logged once per receiver that heard it. Median duplication is six copies in the dense archives. Grouping across receivers destroys cadence and inflates timing features.

Step five uses the true vehicle for splitting and scoring only. Both endpoints of a pair must fall on one side. Grouping on one endpoint lets the other straddle the split. We assert disjointness at run time rather than trusting the splitter.

D. Detector and features

Everything here runs on one desktop machine. The cascade takes about two hours across eight archives. Every other experiment finishes in minutes.

One classifier is used throughout, deliberately. It is a random forest of 200 trees, minimum leaf two. Scikit-learn supplies it, with the seed fixed per run. We do not tune it, because tuning is not the question. A stronger model could shift the absolute values. Whether it would shift the differences we report is untested.

Table III lists the four feature sets. The timing and geometry split carries our central ablation. Readers can therefore contest the partition rather than trust it.

Pair candidates are identities co-observed by one receiver. We compare only claims within one second of each other. A pair is positive when both identities share a true vehicle. The linkage experiment scores 1 588 to 29 218 pairs per archive. Positive rates run from 2.7% to 10.5%.

TABLE III
Feature sets. Pair features describe two identities; identity features describe one.

Set	Members
Pair, geometry	mean, minimum and standard deviation of inter-claim distance; mean absolute speed difference; speed correlation
Pair, timing	presence overlap; comparison count; median nearest-claim gap; phase residual against the beacon period
Identity, rate	message count; duration; median and deviation of inter-arrival
Identity, kinematic	speed mean, deviation and maximum; acceleration mean and deviation; position span in each axis; step mean and deviation; kinematic residual; reported position and heading noise

E. Metrics

We report ROC-AUC and average precision. Both are implemented twice and cross-checked at every call. The first implementation is the normalised Mann–Whitney statistic. The second is scikit-learn, and a run aborts on disagreement. Ties receive mid-ranks, where naive implementations usually diverge. On tied and adversarial inputs the two agree to machine precision.

Three interval kinds appear, and they are not interchangeable. A percentile bootstrap over test rows measures sampling error. DeLong intervals check that bootstrap on the physical-layer result. Spread across random seeds is reported as a subscript instead. Each estimate is printed only beside an interval for itself.

IV. Corpus Reporting Practice

The corpus shows what the literature reports about itself. Of 92 full texts, 22 state an explicit threat model. Nine state a beacon rate and nine an attacker fraction. Four mention a ROC curve, and two any interval or test. Three give a repository link, all to one restricted archive. Eleven papers since 2021 mention ns-2, dormant since 2011. Every count and its pattern ship as a released table.

Both quantities we vary most are the ones least reported. That is the gap this audit measures rather than asserts.

V. Structural Artefacts

Table IV is the complete census. Every column is measured from the raw archives. The scripts producing it import none of our loader.

A. Pseudonyms encode sender identity

Received messages carry both a true sender and a pseudonym. A receiver observes only the pseudonym in deployment. We compared the two fields across every record.

Six archives encode the sender in every identity pair. Both GridSybil archives encode it in 86.5% and 87.0%. Sender 15 emits pseudonyms 10155 and 20155. Sender

TABLE IV
Complete census of the eight VeReMi Extension Sybil archives, measured from source.

Archive	Vehicles	Attacker vehicles	Prevalence (vehicles)	Identities	Attacker identities	Identity prevalence	Encoding rate	Copies per transmission
GridSybil dense	4 064	1 219	30.00%	8 793	5 948	67.6%	86.5%	6
GridSybil sparse	1 688	507	30.04%	3 726	2 545	68.3%	87.0%	4
DataReplaySybil dense	4 067	1 221	30.02%	72 140	69 294	96.1%	100%	7
DataReplaySybil sparse	1 688	507	30.04%	27 421	26 240	95.7%	100%	4
DoSRandomSybil dense	4 067	1 221	30.02%	112 191	109 345	97.5%	100%	7
DoSRandomSybil sparse	1 688	507	30.04%	47 473	46 292	97.5%	100%	4
DoSDisruptiveSybil dense	4 067	1 221	30.02%	112 187	109 341	97.5%	100%	7
DoSDisruptiveSybil sparse	1 688	507	30.04%	47 466	46 285	97.5%	100%	4

Identities per attacker, median and maximum: GridSybil 3/7 dense and 6/7 sparse; DataReplaySybil 56/100 dense and 50/100 sparse; both denial-of-service families 100/100.

TABLE V
Recovering the sender from the pseudonym alone. Decode accuracy names the exact emitting vehicle. Linkage asks whether two pseudonyms share one vehicle.

Measure	Real	Permuted	Chance
Decode accuracy, mean	0.876	0.0031	0.0011
Decode accuracy, range	0.771–1.000	—	—
Linkage AUC, range	0.999–1.000	0.444–0.608	0.5

5853 emits 1058535 and 2058535. The construction resembles $k \cdot 10^n + \text{sender} \cdot 10 + c$.

Two consequences follow. Grouping identities becomes solvable by string arithmetic alone. More subtly, many pipelines aggregate features per true sender. That hands the model the partition it should infer.

The GridSybil shortfall has a single cause. One pseudonym is shared by many distinct senders. Its value is the literal integer one. It carries 147 672 ground-truth messages in the dense archive. Any pipeline keyed on the pseudonym merges those vehicles silently.

B. The pseudonym decodes to its owner

The previous subsection is a claim about string structure. We turn it into a decoder and measure what it recovers.

The decoder is deliberately trivial and untrained. It returns the longest sender identifier found inside the pseudonym.

The control keeps both populations and breaks the correspondence. For decoding, vehicles are permuted among pseudonyms. For linkage, pseudonyms are permuted among identities. A leak shows as a gap between the real and permuted arms.

Table V and Figure 1 report it. The decoder names the emitting vehicle 87.6% of the time. Permuting the correspondence drops that to 0.31%. Chance for these populations is 0.11%.

Linkage from the identifier alone is essentially perfect. Longest common digit substring reaches AUC 0.999 or above. The permuted control sits between 0.44 and 0.61.

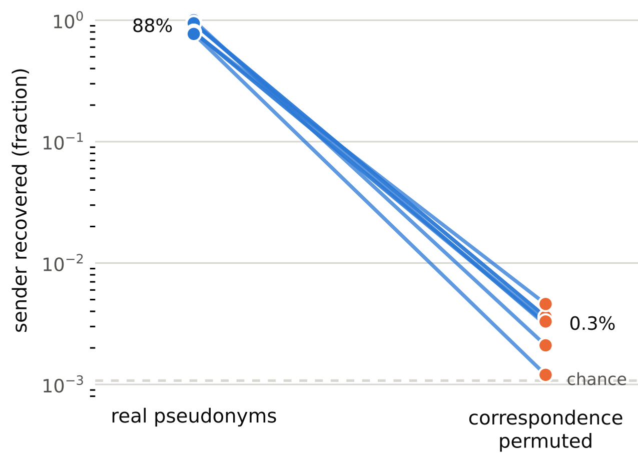

Fig. 1. Recovering the sender from the pseudonym alone. Each line is one archive, on a logarithmic axis. The left point uses real pseudonyms, the right permuted ones. Permuting keeps both populations and breaks only the correspondence. The dashed rule marks chance for these populations.

C. The leak, in bits

Naming the vehicle behind an identity costs 8.1 to 9.5 bits. That entropy is what a detector must resolve from nothing. Let D be the sender named by the untrained decoder above. The residual $H(S \mid D)$ measures 0.00 to 0.75 bits. So $I(D; S)$ recovers 92% to 100% of $H(S)$. The identifier's shape hands over most of the partition.

D. Benign vehicles do not change pseudonyms

We counted identities per vehicle in the eight Sybil archives. Labels came from filenames over the complete member list. Not one benign vehicle emits a second pseudonym. The benign maximum is exactly one across all eight.

F2MD generated these archives and supports pseudonym change policies. Our claim is about the eight Sybil archives we examined. We make no claim about other Extension scenarios or configurations. The policies exist in the generator; they are not exercised here.

One alternative reading would invalidate the measurement. Rotation might occur while the logged sender identifier also changes. The mapping would then be one to one by construction. Renewing the identifier would truncate every observed lifetime. Lifetimes would cluster near the policy period.

They do not. Benign observation spans have median 90.5 to 94.0 seconds. Their coefficient of variation is 0.52 to 0.57. No single value holds more than 1.8% of vehicles. The spread follows radio dwell time, not a fixed schedule. We therefore read the mapping as genuine, not as renaming.

Pseudonym change is absent for honest vehicles in these archives. Unlinkability and pseudonym lifetime cannot be studied on them. Attackers, by contrast, hold many identities each. Identity count alone then separates the classes almost perfectly.

E. Ground truth omits most attacker pseudonyms

This artefact is easy to inherit without noticing. The ground-truth files list one pseudonym per vehicle. That holds for attacker vehicles too, in seven archives. The receiver logs tell a very different story.

For DoSRandomSybil dense the two disagree by 27.6 times. Ground truth names 4 068 identities; the logs show 112 191. The discrepancy is in the identity representation only. Both views contain the same 1 221 attacker vehicles. What ground truth omits is the pseudonyms those attackers emitted. Deriving the Sybil partition from ground truth alone therefore finds nothing. We build labels from receiver logs and filenames instead.

F. Attacker prevalence is fixed and unrealistic

Attacker share is 30.0% of vehicles in every archive. The eight values span 30.00% to 30.04%. Identity-level prevalence is different and much higher. It ranges from 67.6% to 97.5% across the archives.

Our search found no field measurement of V2X prevalence. Thirty percent is not defended anywhere in the documentation. Impact is already severe at 3% in navigation services [30]. We therefore treat prevalence as a variable, not a constant.

G. Attack intensity varies by two orders

Identities per attacker differ sharply between families. GridSybil attackers hold one to seven identities. The other families reach one hundred identities. Papers rarely name the family they evaluated.

H. The preprocessed derivative cannot express the task

One popular derivative offers 2.27 million ready rows. It contains no sender and no pseudonym column. Sybil detection is therefore impossible on it in principle. Nineteen attack types and one benign class are balanced exactly. Row-level attacker prevalence is therefore 95%.

Its receiver column is separately malformed. Just two identifiers account for 95.0% of the rows. The other 113 477 identifiers hold exactly one row each. Receiver-disjoint validation then degenerates without warning. One split in five gave accuracy 1.000 with F1 0.000. Its test fold contained a single class.

TABLE VI
Single-scalar separation, all eight archives. Benign identities beacon at 1.0 s throughout. Brackets give the bootstrap 95% interval.

Archive	AUC interval	AUC count
GridSybil dense	0.657 [.631,.682]	0.635 [.608,.662]
GridSybil sparse	0.576 [.550,.604]	0.601 [.572,.629]
DataReplaySybil dense	0.544 [.531,.557]	0.996 [.992,.999]
DataReplaySybil sparse	0.534 [.520,.549]	0.993 [.989,.997]
DoSRandomSybil dense	1.000 [1.00,1.00]	0.996 [.992,.999]
DoSRandomSybil sparse	1.000 [1.00,1.00]	0.994 [.989,.997]
DoSDisruptiveSybil dense	1.000 [.999,1.00]	0.996 [.992,.999]
DoSDisruptiveSybil sparse	1.000 [.999,1.00]	0.994 [.989,.997]

TABLE VII
Pairwise identity linkage by evidence type.

Family	AUC geometry	AUC timing
GridSybil	0.966–0.979	0.996–1.000
DoSRandomSybil	0.748–0.829	1.000
DoSDisruptiveSybil	0.632–0.716	1.000
DataReplaySybil	no concurrent pairs	—

VI. How Far the Task Is Simplified

A. One scalar solves most archives

We computed two statistics per identity. The first is median inter-arrival time. The second is raw message count. Neither requires training, linkage or physical reasoning.

Direction is fixed before looking at any score. An attacker splits one budget across many ghosts. Each ghost therefore transmits less often, not more. We score larger interval and smaller count as attacker. Table VI gives both statistics per archive.

The mechanism is arithmetic, and we measured its terms. An attacker emits at aggregate rate R across N identities. Each identity then transmits about every N/R seconds. These families run at $R = 2.0\,\mathrm{Hz}$, twice the benign rate. That doubling is the denial-of-service part of the attack. With $N = 100$ the predicted gap is 50 s. We measure 50 s, so arithmetic and archive agree.

Benign identities beacon once per second throughout. Median inter-arrival is therefore a sufficient statistic here. Geometry is redundant by construction, not by weakness. Only the two GridSybil archives resist both trivial statistics.

B. Timing evidence versus geometric evidence

We next linked identity pairs to one vehicle. Splits were vehicle-disjoint and transmissions were deduplicated. Features were partitioned into timing and geometry.

Figure 2 shows the gap per archive. Table VII gives the same result by family. Timing alone reaches unity on every measurable family. Geometry, the physically meaningful signal, is much weaker.

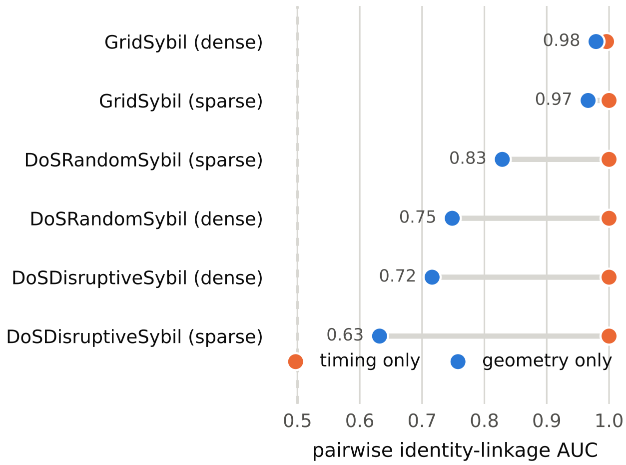

Fig. 2. Timing evidence versus geometric evidence, per archive. Each pair shows one archive under two feature sets. Timing alone sits at or near AUC 1.000 everywhere. Geometry, the physically meaningful signal, varies widely. The dashed rule marks chance at AUC 0.5. DataReplaySybil is absent: it yields no concurrent pairs.

TABLE VIII
Information available to the detector at each stage. GT means a field no receiver can observe.

Stage	GT partition	Raw pseudonym	Native prevalence
S0	yes	yes	yes
S1	yes	no	yes
S2	no	no	yes
S3	no	no	yes
S4–S6	no	no	no

TABLE IX
Protocol cascade. Mean AUC over five seeds. Parentheses give the spread across those seeds. The last column is the untrained scalar on GridSybil dense.

Protocol	GridSybil dense	GridSybil sparse	Other six archives	Trivial scalar
S0 published	0.998 (.000)	0.999 (.001)	1.000	0.009
S1 − pseudonym	0.997 (.000)	0.999 (.001)	≥0.999	0.009
S2 − GT key	0.937 (.002)	0.936 (.001)	≥0.999	0.657
S3 + disjoint	0.931 (.009)	0.934 (.006)	≥0.997	0.674
S4 prevalence 3%	0.893 (.088)	—	≥0.996	0.633
S5 + jitter	0.834 (.131)	—	≥0.999	0.574
S6 − rate feats.	0.843 (.135)	—	1.000	0.574

The mechanism is a fixed transmission schedule. Sibling identities interleave on their attacker's clock. The offset is an exact fraction of the beacon period. Denial-of-service ghosts sit half a second apart. Their residual against half a period has median 3×10^{-6}. GridSybil siblings instead sit one third of a period apart. For unrelated identities the same quantity is 0.25. A round-robin over the transmit slot would explain both offsets. We report that as a reading, not a measurement. The sparse archive does not fit it cleanly. Its attackers hold six identities yet show the same third. Phase alone separates pairs at AUC 0.63 to 0.88. Every one of those separations is significant beyond $p = 10^{-8}$.

An attacker removes this signal by jittering transmissions. A quarter-second jitter cuts phase AUC to 0.55–0.58. On the denial-of-service families that is a fall of 0.27. The change costs nothing in attack utility. Detectors validated on it need not transfer.

Note that DataReplaySybil yields no concurrent pairs. Its identities are used sequentially instead. Pairwise concurrent linkage is therefore inapplicable there.

VII. A Staged Protocol Cascade

Artefacts alone do not prove that published numbers inflate. We therefore ran one detector through seven protocols. Each stage removes exactly one advantage from the previous. The change between adjacent stages is associated with that removal.

The stages are cumulative, so each is nested in the last. An increment therefore depends on what was already removed. Section VII-A tests whether that ordering carries the result.

Stage zero reconstructs common practice in the corpus. Features are aggregated on the ground-truth sender key. The raw pseudonym is offered as a numeric feature. Folds are random and prevalence is the native 30%. Azam et al. describe the aggregation and the folds [23]. The pseudonym feature is our addition, not theirs. We include it to test the encoding artefact directly.

Table VIII states what each stage may see. A real receiver has access only to the last row.

Later stages remove the pseudonym, then the ground-truth key. Then splits become vehicle-disjoint and prevalence drops to 3%. Then transmissions are jittered by up to a quarter second. Finally the message-rate features are dropped entirely.

Table IX reports the result. Six of eight archives stay above AUC 0.995 throughout. No control we apply moves them meaningfully. Only the two GridSybil archives respond to the cascade.

The trivial column reads 0.009 at the first two stages. That is the same statistic scored in the opposite direction. Ground-truth aggregation pools a whole attacker into one row. Pooled attackers beacon faster than honest vehicles, not slower. A receiver cannot know which direction applies.

Removing the pseudonym feature shifts AUC by 0.0006. The feature keeps the raw representation it has in the archive. The identifier adds nothing once aggregation is already leaking.

Removing the ground-truth key is followed by falls of 0.060 and 0.064. This is the largest and only unambiguous change. Both are far outside the seed spread at that stage. Making the split disjoint is followed by a further 0.006. Inferring the partition, rather than receiving it, is the task.

Below this point the archive stops supporting inference. At 3% prevalence the seed spread reaches 0.088. The two sparse archives become degenerate and are dropped. Later stage differences all sit inside that spread. We therefore do not attribute them to the controls. Section VIII measures those knobs where estimates are stable.

TABLE X
Each advantage added back to the controlled protocol. Baseline AUC 0.843. Paired intervals are on shared rows, one seed.

Advantage added	AUC	Δ	Δ paired, one seed
ground-truth key	0.999	+0.157	not pairable
random folds	0.923	+0.080	not pairable
native prevalence	0.916	+0.074	not pairable
no jitter	0.884	+0.042	not pairable
pseudonym feature	0.859	+0.016	+0.040 [-0.013, 0.140]
rate features	0.833	-0.009	+0.047 [-0.004, 0.140]

The final column carries the uncomfortable comparison. It shows GridSybil dense, where the model beats the scalar. The margin is 0.28 on the dense archive and 0.36 on the sparse one. On the four denial-of-service archives that margin vanishes. There the untrained scalar itself reaches 0.999 from stage S2. The learned model adds nothing at all on those archives. GridSybil is the one family where the benchmark measures something.

A. The ordering does not carry the result

A cumulative cascade cannot separate a factor from its predecessors. We therefore measure each advantage against one fixed baseline. Starting fully controlled, we add exactly one advantage back. Every effect is measured against the same reference. These effects carry no ordering by construction. Table X reports them.

Two factors change which rows exist and cannot be paired. Ground-truth aggregation scores vehicles, not identities. Prevalence subsampling changes the population scored. For those we report the change without a paired interval.

The two delta columns can disagree in sign, and do twice. They measure different things and neither is wrong. The mean column averages five splits, each scored separately. The paired column is one split, scored on shared rows.

At 3% prevalence a split holds about six attackers. One split therefore moves easily against the five-split mean. The paired intervals span 0.15, wider than either delta. Both intervals include zero, so neither factor is resolved. We report both columns rather than the one we prefer.

The two designs agree on what matters. Ground-truth aggregation is the largest single advantage. At +0.157 it is nearly twice the next largest factor. The forward cascade ranked it first as well. Ordering is therefore not producing that conclusion.

They disagree on the smaller factors, which is expected. Random folds cost little at native prevalence with rate features. They cost more once those advantages are already gone. That is interaction, and this design exposes it.

The pseudonym feature remains the smallest advantage. Its paired interval includes zero on the shared rows. This is the third design in which it adds nothing.

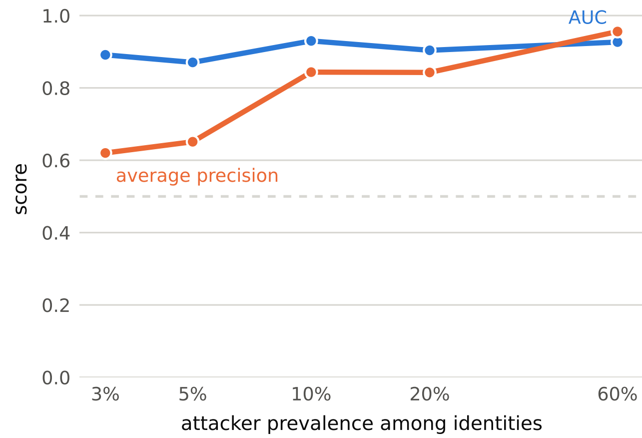

Fig. 3. The two metrics respond differently to prevalence. Ranking is nearly flat; average precision is not. A result reported at native prevalence does not transfer. GridSybil dense, three seeds per point.

B. An earlier version of this test was wrong

Our first attempt reported no inflation at all, and was invalid. It offered the hashed identity token as the pseudonym feature. Hashing had already destroyed the sender encoding. The naive arm therefore never carried the leak it claimed. We recovered the true pseudonyms and repeated the comparison. The corrected conclusion is narrower but better supported.

VIII. Sensitivity

Papers rarely report the four knobs that matter most. We sweep each one and hold the others fixed. Table XI collects every sweep. Sweeps use GridSybil dense, the family that resists the shortcuts. Figure 3 shows the prevalence contrast.

Prevalence is the knob papers omit most often. These archives set a floor on how low it can go. Subsampling to prevalence p holds the benign pool fixed. Surviving attackers are $p\,N_b/(1-p)$, which shrinks fast. GridSybil dense keeps 15 attacker identities at 2%. That is below the twenty we require before splitting. At 3% it keeps 22, of which about 6 reach the test fold. The measured floor is 3% dense and 5% sparse. The binding constraint is the benign population, not the attackers. Only 718 benign identities exist in the dense archive.

Between 3% and native the ranking metric barely moves. AUC stays between 0.871 and 0.930 across that range. Average precision behaves completely differently. It falls from 0.956 at native to 0.620 at 3%. Precision, recall and F1 inherit that dependence directly. A paper reporting them at 30% reports a different problem.

Jitter costs little on the per-identity task. Half a second of jitter moves AUC from 0.927 to 0.916. The per-identity rate gap survives jitter of that size. The pairwise clock signal does not, as shown next.

Attack intensity matters less than expected. Capping attackers at two identities gives AUC 0.910. Leaving them uncapped gives 0.927.

TABLE XI
Sensitivity on GridSybil dense. Three seeds per point. Prevalence uses a disjoint split; the other sweeps run at native prevalence.

Setting	AUC	seed sd	AP	Trivial
attacker prevalence				
0.005	degenerate: 4 attackers kept			
0.010	degenerate: 7 attackers kept			
0.020	degenerate: 15 attackers kept			
0.03	0.891	0.074	0.620	0.573
0.05	0.871	0.091	0.651	0.599
0.1	0.930	0.018	0.844	0.648
0.2	0.904	0.014	0.843	0.628
transmission jitter (s)				
0.0	0.927	0.007	0.956	0.671
0.05	0.928	0.007	0.959	0.686
0.1	0.928	0.007	0.958	0.686
0.25	0.922	0.009	0.953	0.686
0.5	0.916	0.018	0.947	0.686
identities per attacker				
2	0.910	0.004	0.911	0.707
3	0.916	0.009	0.922	0.681
5	0.925	0.006	0.950	0.670
10	0.927	0.007	0.956	0.671
25	0.927	0.007	0.956	0.671
50	0.927	0.007	0.956	0.671
100	0.927	0.007	0.956	0.671
native	0.927	0.007	0.956	0.671
fraction of beacons kept				
1.0	0.927	0.007	0.956	0.671
0.75	0.914	0.013	0.951	0.659
0.5	0.906	0.008	0.946	0.612
0.25	0.909	0.010	0.951	0.598
0.1	0.897	0.003	0.946	0.559

Beacon rate has a clearer effect. Keeping one beacon in ten drops AUC to 0.897. The trivial baseline falls further, from 0.671 to 0.559. Sparse beaconing hurts the shortcut more than the model.

IX. VeReMi 2018 as a Physical-Layer Substrate

Signal-strength methods appear in 33 indexed records. They need received power, which the Extension omits. VeReMi 2018 does contain populated received power. It contains no Sybil families, only position falsification. Neither VeReMi release therefore carries both together.

One dataset outside the VeReMi family does carry both. Guven and Tayşi built it for exactly this gap [29]. It pairs simulated RSSI with several Sybil attack variants. We could not evaluate it here. Its repository carries an owner takedown and authorship-dispute notice. All data and code were restricted when we sought access. We report that as a fact about availability, not about quality.

We fitted log-distance path loss over 86 919 links. Benign claims are honest, so their distances are trustworthy. Table XII reports the benign-only and all-link fits.

Nine decibels per decade is very shallow. Median per-link scatter is another 3.6 decibels. A tenfold position lie barely clears that noise.

We then implemented position verification properly. Receivers were placed from their own position reports. Residuals were summarised per link and classified. Senders were partitioned into five folds, disjoint by construction. Every link is therefore scored once, by an unseen model.

TABLE XII
Path-loss behaviour and detector performance, VeReMi 2018. Brackets are 95% intervals for the estimator on that row. Free space would give -20 dB/decade and exponent 2.0. Chance is AUC 0.5, and the attacker base rate is 0.055.

Quantity	Benign fit	All links
Correlation, $\log_{10} d$	-0.639	-0.633
Slope (dB/decade)	-9.41	-9.29
Path-loss exponent	0.941	0.929
Scatter, median (dB)	3.56	3.57
AUC across folds	0.524 [.502,.547]	0.530 [.507,.553]
AUC out of fold	0.493 [.485,.501]	0.499 [.491,.507]
Average precision	0.054	0.054

Two estimators are reported, each with its own interval. Averaging AUC across the five folds gives 0.530. Its interval across folds runs 0.507 to 0.553. Pooling the out-of-fold scores instead gives one global ranking. That ranking reaches 0.499, with interval 0.491 to 0.507. DeLong and bootstrap intervals agree to three decimals. Average precision is 0.054 against a 5.5% base rate.

The gap between the two estimators is itself informative. Fold-local ranking is weak; global ranking is indistinguishable from chance. A deployed detector needs one threshold, not five. Raw signal statistics without geometry reach 0.597. That figure averages five sender-disjoint random splits. It is a fold-average, so it compares with 0.530, not 0.499.

Refitting on benign links alone changed almost nothing. The exponent moved from 0.929 to 0.941. Fold-mean AUC moved from 0.530 to 0.524. The conclusion is therefore not an artefact of that choice.

The attack tested is constant position falsification. It is the most blatant lie available, and goes undetected here. We therefore make a narrow claim about this substrate. VeReMi 2018 cannot meaningfully validate signal-strength position verification. Whether the method works on calibrated data is untested here.

X. Do the Shortcuts Generalise?

Every result so far comes from one dataset family. The obvious objection is that these are quirks of one artefact. VeReMi NextGen is the right test of that.

It uses InTAS rather than LuST, a later generator, another attack. It was built to address limitations of the earlier releases. We ran the same measurements on its four scenarios.

Table XIII separates what was fixed from what was not.

The identity leak is genuinely repaired. The receiver-visible alias no longer encodes the sender. Linkage from it sits at chance in all four scenarios. The artefact we weight most heavily is gone.

Three things survive the redesign. Benign identities still never rotate, so privacy stays unstudiable. Receiver duplication persists, at a median of 2 to 49 copies. Message count alone still reaches AUC 0.972 to 0.996.

TABLE XIII
The same measurements on VeReMi NextGen, four scenarios. Alias is what a receiver sees; sender identifier is ground truth.

Measurement	Extension	NextGen
Alias encodes the sender	86.5–100%	0.0%
Linkage AUC from the alias	0.999–1.000	0.494–0.506
Benign identities that rotate	0	0
Message-count AUC, untrained	0.993–0.996	0.972–0.996
Median copies per transmission	4–7	2–49
AUC from the sender identifier	—	1.000

A new identifier artefact replaces the old one. Sybil identities carry synthetic sender identifiers. Benign identifiers reach 32 564 across the four scenarios. Every attacker identifier exceeds 1 000 078. The two ranges are disjoint in every scenario. The identifier alone therefore separates the classes perfectly.

This field is ground truth, so no receiver can use it. It matters because published pipelines do key on that field. That is stage S0 of our own cascade, reconstructed from practice. Any pipeline touching it inherits a perfect shortcut.

One caveat limits what we can compare. In NextGen a ghost carries its own sender identifier. That field therefore names an identity, not a vehicle. Vehicle-level prevalence is not recoverable from it. We report identity-level figures only, and say so.

XI. Discussion

A. What should change

We propose the cascade itself as a pre-publication check. A result surviving S0 to S6 has earned its number. A result collapsing at S2 was reading the partition. The released script accepts any detector through a model flag. A logistic regression behaves the same way, moving 0.096 against 0.064. The ladder is a property of the data, not of one model.

Seven reporting rules follow from the measurements. Always state the attacker fraction, beacon rate and density. Name the attack family, and separate timing from geometry. Compare against a single-scalar baseline before any model. Match evidence per identity when comparing policies. Use identity-free tokens and vehicle-disjoint splits. Publish the prevalence sweep, not one operating point.

B. What our own errors taught us

Nine defects were found and corrected during this work. Duplicate observations inflated every timing feature initially. Vehicle identifiers repeat across independent simulation runs. One comparison pitted a pooled view against a single observer. A sampler emptied the positive class at large scale. A fixed output filename silently overwrote earlier results. A leakage feature was hashed before being offered as leakage. A path-loss model was fitted on the links it should exclude. A point estimate was printed beside an interval for another estimator. A reported table was published with no script behind it.

One diagnostic recurred each time: a result too strong for its signal. No pair feature passed AUC 0.65, yet a forest on them reported 1.000. That gap indicated leakage, and leakage was present.

XII. Limitations

Our audit covers the Sybil archives specifically. Other attack families may behave differently.

Simulation realism bounds everything we report. These archives model propagation and mobility, not radios. A shortcut seen in simulation need not survive on hardware. Equally, hardware may add shortcuts a simulator never shows.

Jitter is applied afterwards to logged transmission times. It ignores medium access, so it bounds how cheap the escape is. We report one uniform distribution, not a channel model.

Observations are not independent within a vehicle. Vehicle-disjoint splits handle the worst of that dependence. Temporal dependence within a fold remains, and inflates precision.

Two further limits are structural. Emulated infrastructure uses claimed positions for reception. That choice favours the infrastructure observer. The prevalence floor is 3% dense and 5% sparse. That limit is a property of the data, not our method.

External validity rests on two datasets, not one. Both remain simulator output from one research lineage.

XIII. Conclusion

We measured what the VeReMi benchmarks contain. Pseudonyms encode sender identity almost everywhere. Honest vehicles never change pseudonyms in these archives. Attacker prevalence is fixed at thirty percent throughout. Ground-truth files omit most pseudonyms the attackers emitted. One untrained scalar solves six of eight archives. Timing alone reaches near-perfect linkage accuracy. Geometry, the meaningful signal, performs far worse.

We then staged the protocol from published practice to deployment. Removing ground-truth aggregation moved the score furthest. The pseudonym feature added nothing once aggregation was fixed. Ranking metrics survive a prevalence change; precision metrics do not. Finally, VeReMi 2018 cannot validate the signal-strength family. An accessible calibrated dataset is therefore a necessity.

Our code, controls and checklist accompany this paper. We hope future results become comparable again.

Ethics

This work analyses published artefacts and involves no human subjects. The archives are simulated, so no vehicle or driver is real. We name published work to locate protocol choices, alleging no misconduct. One dataset we describe is withdrawn under an authorship dispute. We report its stated availability, take no position, and never obtained it.

Availability

Scripts, controls and result tables are released with this paper. They reproduce every number above from the raw archives. The archive is deposited at 10.5281/zenodo.22397725. A container image rebuilds and re-verifies it in one command. The full eight-archive census ships as a machine-readable table. Every statistic we measured is a column, not only those printed here.

Every result file names the script that produced it. A manifest records a hash per file and the library versions. The archive states the route from raw data to every table. Independent metric implementations are included and self-tested.

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